MIL-D-18576A(SHIPS)
25 August 1961
SUPERSEDING
MIL-D-18576(SHIPS)
30 March 1955

MILITARY SPECIFICATION

DUMMY, LOAD, ELECTRICAL,

NAVY TYPE 14ACN

1. SCOPE

1.1 This specification covers the Dummy Load Navy type 14ACN. This test equipment shall be capable of dissipating radar-modulator pulse energy and shall also serve as a voltage divider for test purposes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-L-3661 - Lampholders and Lights, Indicator; Bayonet Base, Miniature and Candelabra
MIL-L-15098 - Lamps, Glow.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MS35059 - Switches, Toggle, Double Pole, One-Hole Mounting (Sealed Toggle Bushing).

DRAWINGS

BUREAU OF SHIPS

REA49131 - MX-913/U Cap and Chain, for Use With Type "N" Connectors.
REB49252 - Receptacle Connector, Navy Type 49194.
REB49312 - Receptacle Connector UG-37A/U.

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - A preproduction sample shall be furnished for the tests specified in 4.2.

3.2 Material. - The material entering into the construction of the dummy load shall be such that it shall meet the electrical, mechanical and environmental requirements of this specification.

3.3 General requirements. - During all phases of design of the dummy load, prime consideration shall be given to reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service.

3.4 External finish. - The external finish of the dummy load shall be gray enamel in accordance with Specification MIL-E-16400.

3.5 Dimensions and weight. - The overall dimensions of the dummy load shall not exceed 26 by 13-1/4 by 11-3/4 inches. The maximum weight of the dummy load, including the shockmounting plate, shall not exceed 44 pounds. The form factor shall be as shown on figure 1.

3.6 Shockmount. - The dummy load shall be shockmounted to a plate by four vibration mounts conforming to Lord type number 200 ph-15, or equal.

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3.7 Power dissipation network. - The power dissipation network shall consist of twelve 600 ohms noninductive resistors in parallel. Each resistor shall have a nominal power rating of 150 watts (see figure 2).

3.8 Impedance. - The dummy load shall provide a 50 ± 2-ohm termination for the modulator of the radar set.

3.9 Fan motor. - The fan motor shall operate from a 115-volt, 60-cycle, single-phase external supply. Provisions shall be made in the fan circuit for the following:

(a) Lamp, neon, type NE51 in accordance with Specification MIL-L-15098.
(b) "ON-None-Off" toggle switch in accordance with Standard MS35059.
(c) Indicator light in accordance with Specification MIL-E-3661.
(d) Fuseholder, extractor, front mounted type, Bussmann Type HKP or equal.
(e) Fuse, 1-1/4 by 1/4 ferrule type, 1-1/2 Amp 250V., 3AG type.
(f) Air filter shall be provided for the blower.

3.10 Voltage divider. - The voltage divider shall consist of two capacitors; one variable, 2 to 4 micromicrofarads ($\mu\mu$f), the other fixed at 40$\mu\mu$f, between the input and ground (see figure 2). The capacity voltage divider shall provide a means to observe the modulator pulse applied to the dummy load. The voltage divider shall divide the input voltages up to 10,000 volts peak, in the ratios of 20 to 1 or 10 to 1, dependent upon the setting of the variable capacitor. The voltage available at the end of a 5-foot piece of 50-ohm Radio Frequency Cable RG-214/U (30$\mu\mu$f per foot) shall be between 100 to 200 volts peak when 10,000 volts peak is applied at the input of the dummy load.

3.11 Enclosure. - The equipment shall be enclosed in a cover designed as an integral part of the dummy load. This cover shall form an instrument case for the dummy load.

3.12 Input connector. - The power input connector shall be a pulse Receptacle UG-37A/U as shown on Drawing REB49312 and shall be fitted with a protective cap secured by a 6-inch length of beaded chain.

3.13 Output connector. - The output connector for pulse monitoring purposes shall be a Navy type 49194 as shown on Drawing REB49252 fitted with Cap MX-913/U as shown on Drawing REA49131.

3.14 Frequency range. - The dummy load shall operate over the frequency range of 0 to 4.5 megacycles per second.

3.15 Power dissipation. - The dummy load shall withstand 5 cycles at 3 kilowatts average power with the fan operating and five cycles of 1.8 kilowatts average power with the fan operating in accordance with the test cycling procedures of 4.5.3. The load shall dissipate the power specified herein with voltage pulses up to 10,000 volts peak. There shall be no voltage breakdown and no evidence of deterioration or excessive heating.

3.16 Environmental requirements. -

3.16.1 Shock and vibration. - The dummy load shall be designed to withstand the shock and vibration test specified in 4.5.4.

3.16.2 Humidity resistance. - The dummy load shall be designed to withstand the humidity test specified in 4.5.5.

3.16.3 Requirements after environmental tests. - After the environmental tests, the dummy load shall show no evidence of mechanical or electrical failures. There shall be no signs of peeling or flaking of the painted finish, or corroding of the metallic parts.

3.17 Identification plate and marking. - The dummy load shall be marked for identification in accordance with Standard MIL-STD-130. The marking shall be applied to an identification plate which shall be firmly attached to the front panel, and shall include the following:

(a) Nomenclature.
(b) Manufacturer's identification.
(c) Contract or order number.
(d) Special characteristics.

3.17.1 Each dummy load shall be marked with the following: "Danger - High Voltage".

3.17.2 Technical manuals. - Two technical manuals shall be furnished with each dummy load.

3.17.3 Assignment of nomenclature. - The contractor shall request the assignment of nomenclature from the bureau or agency concerned.

3.18 Workmanship. - Workmanship shall be such that the finished product will meet all requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be

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kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - One preproduction sample shall be made and assembled with tools and methods, that, as far as practicable, are the same as those which will be used for quantity production of the product. Preproduction inspection may be waived if the manufacturer has written approval, which has been in existence for less than 18 months from the bureau or agency concerned. The manufacture of the dummy loads for a Government contract, prior to approval of the preproduction sample, shall be at the manufacturer's risk. The preproduction samples shall be subjected to the following examinations and tests. No failures shall be permitted.

Examination or test	Requirement paragraph	Test paragraph
Visual, dimensional and mechanical examination	3.3 through 3.10, and 3.17 through 3.18	4.5.2
Power dissipation	3.15	4.5.3
Humidity resistance	3.16.2	4.5.5
Shock and vibration	3.16.1	4.5.4

4.3 Acceptance inspection tests. - A sample number of equipments shall be selected in accordance with 4.4.2, 4.4.3 and 4.4.4 and each of the selected dummy loads shall be subjected to the following tests. The tests shall be performed in general in the order listed.

Test	Requirement paragraph	Test paragraph
Group A		
Visual, dimensional, and mechanical examination	3.3 through 3.10 and 3.17 through 3.18	4.5.2
Group B		
Power dissipation	3.15	4.5.3
Group C		
Humidity resistance	3.16.2	4.5.5
Shock and vibration	3.16.1	4.5.4

4.4 Sampling and acceptance inspection. -

4.4.1 Lot. - All dummy loads offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.4.2 Sampling for group A tests. - Sample dummy loads shall by selected in accordance with Standard MIL-STD-105 at inspection level II for the tests specified in 4.3. The Acceptable Quality Level (AQL) shall be 1.5 percent.

4.4.3 Sampling for group B tests. - Sample dummy loads shall be selected in accordance with level L-6 of the table for small sample inspection of Standard MIL-STD-105. The AQL shall be 1.0 percent.

4.4.4 Sampling for group C tests. - Group C tests shall be performed when specified by the bureau or agency concerned. One complete dummy load shall be selected and subjected to each of the group C tests specified in 4.3 with acceptance based on 4.4.5.

4.4.5 Acceptance, group A, B and C tests. - The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.5 Test procedures. -

4.5.1 Test equipment and inspection facilities. - Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.5.2 Visual, dimensional, and mechanical examination. - Dummy loads shall be examined visually, dimensionally, and mechanically to determine conformance with this specification.

4.5.3 Power dissipation. - The specified average power (see 3.15) shall be applied at any frequency within the specified range (see 3.14). Thermal stability shall be reached prior to the power dissipation cycling test. The dummy load shall withstand the cycling test specified in 3.15 for a period not less than 15 minutes with the equipment on and 15 minutes with the equipment off.

4.5.4 Shock and vibration. - The dummy load shall be tested while mounted on a standard mounting fixture for deck- or platform-mounted parts in accordance with method 207 of Standard MIL-STD-202. The equipment shall then be firmly mounted on the vibration platform and tested in accordance with method 201A of Standard MIL-STD-242. After the shock and vibration tests have been completed, the dummy load shall meet the requirements of 3.16.1.

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4.5.5 Humidity resistance. - The dummy load shall be subjected to the humidity test in accordance with method 106A, with the exception of step 7, of Standard MIL-STD-202.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Dummy load. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking. - Dummy loads shall be preserved and packaged by level A or C; packed by level A or B and marked in accordance with the requirements of Specification MIL-E-17555. Method II shall apply for level A preservation and packaging.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 5985-N129Sh)

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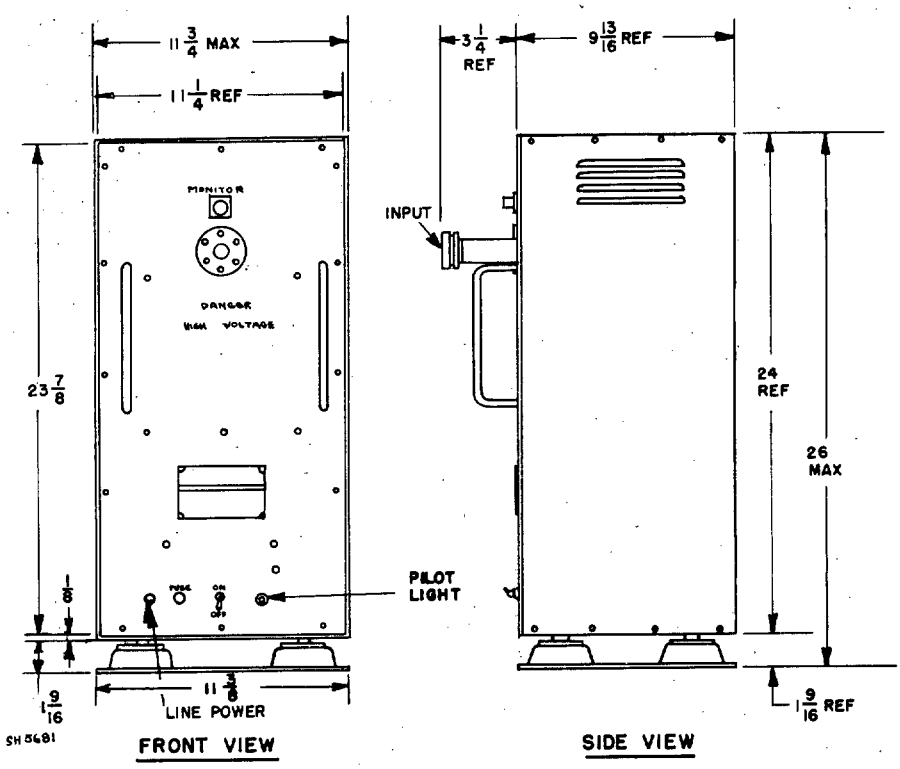


Figure 1 - Dimensional outline of Dummy Load 14ACN.

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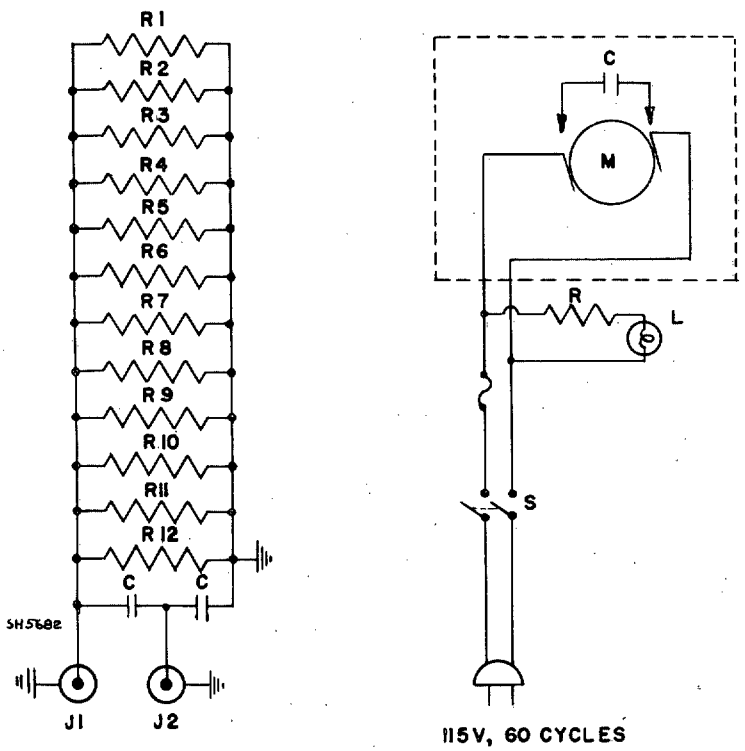


Figure 2 - Wiring diagram.

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No
   If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.)
Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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